CS F364: Design & Analysis of Algorithms

Quiz 5 — June 12, 2026

Coverage: Lectures 1–14

> **Note**
>
> **Instructions:**
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> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question: LCS — Dynamic Programming

Let $c[i,j]$ be the length of the LCS of prefixes $X[1..i]$ and $Y[1..j]$.

For strings $X =$ `ABCD` and $Y =$ `AXCD`:

(a) [2 marks] Write the complete recurrence relation for $c[i,j]$ for all $i, j > 0$. Clearly state the base case and both cases ($x_i = y_j$ and $x_i \neq y_j$).

(b) [2 marks] Fill in the DP table below. The first row ($i = 0$) and first column ($j = 0$) are already given.

$c[i,j]$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
	$\emptyset$	**A**	**X**	**C**	**D**
$i = 0$ $\emptyset$	0	0	0	0	0
$i = 1$ **A**	0				
$i = 2$ **B**	0				
$i = 3$ **C**	0				
$i = 4$ **D**	0				

(c) [1 mark] What is the length of the LCS? What is the actual LCS sequence?